Eya Cherif

Data Scientist

Leipzig, Germany | Portfolio | LinkedIn | GitHub

Profile Summary

Computer scientist specializing in data science and machine learning, with five years of experience working on deep learning applications. Experienced in transforming remote sensing data into production-ready solutions across environmental domains. Enthusiastic and motivated to continuously learn new technologies, with certifications from Esri, Microsoft Azure (AZ-900), DeepLearning.AI, and Ironhack.

Experience

Jan 2026 – Mar 2026
Remote

Ironhack Germany GmbH

Full-time upskilling program focused on full-stack Machine Learning (ML) engineering

Selected projects:

- Built an end-to-end deep learning pipeline for tree crown segmentation from aerial imagery, enabling automated geospatial analysis
- Developed a Retrieval-Augmented Generation (RAG) dashboard using Large Language Models (LLMs) to generate automated report
- Built a Transformer-based Natural Language Processing (NLP) model for fake news detection
- Developed a LangChain-based agent integrating OpenAI APIs for transcription, translation, and web search
- Developed and evaluated machine learning models (Computer Vision, Regression, Classification, Unsupervised methods) on real-world datasets through Kaggle challenges
- Conducted exploratory data analysis (EDA) and built a data preparation pipeline for global forest cover loss data to generate insights supporting environmental monitoring

Python, PyTorch, Scikit-learn, XGBoost, Streamlit, LangChain, SQL, Git, Rasterio, NLTK, Tableau

Jun 2021 – Dec 2025
Leipzig

Scalable Data Analytics and Artificial Intelligence Center (ScaDS.AI) and Remote Sensing for Earth System Research (RSC4Earth)

ML Researcher

- Built an end-to-end data pipeline for dataset curation, preprocessing, and quality control, enabling scalable and reproducible ML workflows
- Developed and optimized multi-task deep learning models (1D-EfficientNet, Transformers, Masked Autoencoders (MAE), Generative Adversarial Networks (GANs)) for vegetation parameter prediction with results published in Remote Sensing of Environment and Biogeosciences
- Evaluated model performance using quantitative metrics, comparing results against state-of-the-art models
- Built an interactive application to visualize spatial predictions and uncertainty maps from satellite imagery
- Presented results at international conferences

Python, PyTorch, TensorFlow/Keras, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Rasterio, QGIS, Google Earth Engine (GEE), Gradio, FAISS (Facebook AI Similarity Search), Git

Feb 2024 – Jun 2024
Montreal, Canada

Mila, Quebec AI Institute and Montreal University (UdeM)

Research Intern

- Curated a diverse biodiversity dataset and developed self- and semi-supervised learning models to advance ecological monitoring, leading to a publication at NeurIPS 2025.
- Trained and evaluated deep learning models on High-Performance Computing (HPC) infrastructure, accelerating experimentation and model evaluation
- Implemented MLOps best practices including modular pipelines, configuration management, and experiment tracking, improving workflow reproducibility

Slurm, Bash/CLI, Python, PyTorch, Git, Weights & Biases

May 2020 – Dec 2020
Kranzberg

Esri Deutschland GmbH

Master's thesis

- Preprocessed multi-temporal Synthetic Aperture Radar (SAR; Sentinel-1) and multispectral (Sentinel-2) imagery to generate analysis-ready datasets for model training
- Developed UNet, DeepLab and a novel DeepForest segmentation architecture for Land Use and Land Cover (LULC) classification, achieving results published in Remote Sensing (2022)

- Integrated the LULC classification tool into ArcGIS Pro for operational use, enabling visualization and practical use of geospatial predictions

SNAP, ArcGIS Pro, Python, TensorFlow/Keras, HPC

Nov 2018 – May 2019 **Atos Information Technology GmbH**
Munich Atos IT Challenge 2019: 8th edition **Machine Learning for Sustainability**

- Designed and developed a chatbot-based recommendation system for product re-use, reaching the final stage of the Atos IT Challenge 2019
- Integrated YOLO object detection to recognize products in real time, enabling personalized recommendations within the chatbot interface

Python, YOLO, Dialogflow, Association Rules

Jun 2018 – Aug 2018 **Standard Sharing Software (3S)**
Tunis, Tunisia Engineering Intern

- Developed a smart parking platform, improving real-time parking management

MEAN Stack, NativeScript, Git, MongoDB

Education

Jun 2021 – Present **PhD candidate in Physics and Earth sciences, Leipzig University**
Leipzig Research topic: Deep learning for vegetation parameter retrieval from hyperspectral data

Oct 2018 – Jan 2021 **Master of Science in Computer Science, University of Passau**
Passau Selected modules: Advanced Topics in Data Science, Deep Learning, Visual Analytics, Database Technologies, Big Data Technologies

Sept 2016 – Jan 2021 **ICT Engineering, Higher School of Communication of Tunis (SUPCOM)**
Tunis, Tunisia Selected modules: SQL, C/C++, Java, MATLAB, Signal Processing, Entrepreneurial Project, Mobile development, Geomatics

Sept 2014 – Jun 2016 **Pre-engineering Studies (Mathematics-Physics), University of Tunis El Manar**
Tunis, Tunisia Rank: 143 / 2,280 (National Engineering Entrance Exam)

Skills

Tools

- **Machine Learning / Deep Learning:** Supervised, Unsupervised, Self-supervised Learning, Scikit-learn, XGBoost, SciPy, PyTorch, TensorFlow/Keras (Coursera certified), Transformers, GANs, UNet, DeepLab
- **Geospatial & Remote Sensing:** Rasterio, GDAL, GeoPandas, xarray, SNAP, QGIS, ArcGIS Pro (Esri-certified), GEE, laspy, Multi- Hyperspectral imagery, SAR (Synthetic Aperture Radar), LiDAR ()
- **Computer Vision & NLP**: OpenCV, Albumentations, NLTK, Word2Vec, GloVe
- **Generative AI & LLMs:** LangChain, Retrieval-Augmented Generation (RAG), OpenAI API
- **MLOps & Cloud:** Docker, Slurm, AWS EC2, Microsoft Azure (AZ-900), Weights & Biases, CI/CD
- **Programming & Dev:** Python, Bash/CLI, MATLAB, C/C++, Git
- **Data & Visualization:** Pandas, NumPy, Matplotlib, Seaborn, Plotly, Streamlit, Gradio, Folium, Tableau
- **Databases:** SQL, MySQL, MongoDB, Neo4j, Firebase, CouchDB
- **Project Management:** Kanban

Languages

Arabic – C2 (Native) | French – C1 | English – B2 | German – B2

Awards & Achievements

- Published at Neural Information Processing Systems (NeurIPS; 2025, USA)
- Mila Research Scholarship (2024, Canada)
- Top 5 Finalist, EARSeL Young Scientist Award (2022, Germany)
- DAAD STIBET Scholarship (2019, Germany)
- Finalist at Atos IT Challenge 2019 (2019, Germany)
- Erasmus+ Scholarship (2018, Germany)

Arts & Music

Music Diploma | Member of international band (2019-2021, University of Passau) | Chorbeau German-French Choir (since 2022, Leipzig)